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Dissertation/Project Coversheet

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Abstract:

This dissertation develops and evaluates a human-centred decision support system (DSS) for evidence-based player selection in European football. The DSS was implemented as a Streamlit application using a season-level Kaggle dataset. The pipeline retrieves player information by league, club, position and minimum minutes played, so positional performance metrics were selected (goals and xG for forwards; passes and xAG for midfielders). Three weighting techniques were applied: Random Forest (feature importance for predicting goals), Entropy (information dispersion) and CRITIC (standard deviation and inter-criterion correlation. Weights were then applied in the TOPSIS algorithm to generate positional-specific lists. Ranking quality was evaluated directly; therefore, Normalised Discounted Cumulative Gain (NDCG) assesses how well TOPSIS ranking correlates with graded relevance when ground-truth or proxy relevance scores are available, and Kendall's τ assesses agreement and consistency between rankings produced by different weighting schemes or against reference lists; these metrics are computed on a per-position basis and aggregated to allow sensitivity analysis. Results are presented via interactive tables and visualisations and can be exported for further use. The primary limitation is the absence of a market-value variable in the dataset, so economic analyses cannot be conducted and are recommended for future work. This dissertation offers a fully transparent, replicable Multi-criteria decision making (MCDM) methodology and actionable recommendations for evidence-based team development in professional football.

Key terms: *Player selection; Multi-criteria decision making (MCDM); TOPSIS; Random Forest; Entropy; CRITIC; Football analytics*.

1. Introduction

1.1. Background

Football is the most popular and lucrative sport in the world, and it has significant economic and cultural impacts far beyond the pitch (Ben Jeddou, 2024). In Europe, elite clubs such as Real Madrid and AC Milan operate not only as sports teams but as complex organisations with strong managerial and financial influence. Their success depends on striking a balance between sporting results and sound financial sustainability. Behind the passion, football has evolved into a highly competitive industry in which effective management is as decisive as player performance.

Sporting directors and other executives have an important role to play, and they are responsible for creating teams that are competitive on the pitch while also complying with Financial Fair Play regulations (Galaz-Cares, 2024). Globalisation and digital transformation have increased competition for top talent and raised transfer costs; therefore, it is crucial to have decision-support tools (such as attribute selection, ranking, and list generation for each position) rather than relying on predictions of future outcomes. There is a need for such decision-support tools, which has been highlighted in research on player development and transfer fees (van Arem, Goes-Smit and Söhl, 2025), because machine-learning techniques are increasingly used to extract valuable insights from large volumes of match and season data (Markopoulou, Papageorgiou and Tjortjis, 2024), and comparative rating systems provide an objective method of ranking players and aiding selection (Arntzen and Hvattum, 2020), so they are essential tools for sporting directors and other executives.

1.2. Problem Statement

Despite the availability of new technologies such as business intelligence and machine learning (ML), football transfers remain highly speculative and risky, and acquiring players represents a significant investment for clubs, which can have long-term negative consequences if poor decisions are made (Follert and Gleißner, 2024). Traditional scouting techniques rely heavily on subjective judgments and past performance, which may not accurately reflect future performance, because they are based on incomplete information.

Furthermore, the increase in transfers due to regulations such as the Bosman ruling has further complicated the situation for smaller clubs, exposing them to greater risk (Galaz-Cares, 2024), thus making it more challenging for them to compete with larger clubs. Transfer decisions are influenced by numerous interrelated variables, including player performance, cost, age, injury history, and potential resale value, and therefore require a comprehensive approach to evaluation. The literature highlights a lack of effective frameworks that integrate machine learning, multi-criteria decision-making models and management principles to support transfer decisions. To address this gap, this dissertation develops a practical, data-driven decision-support framework that systematically evaluates and ranks players using season-level historical performance metrics; the tool is explicitly intended to complement and inform human scouting rather than replace scouting judgement.

1.3. Relevance of the Study

This research contributes in two ways. Firstly, it advances the literature on applying decision science, artificial intelligence (AI) and management theory in sport (Markopoulou, Papageorgiou and Tjortjis, 2024) by using objective weighting and MCDM with a transparent ranking process, moving beyond purely descriptive approaches to provide actionable recommendations for player selection.

Secondly, the dissertation develops a decision-support tool for player selection that is not intended to replace scouts and managers but to assist them by providing position-specific ranked lists and comparisons across alternative weighting schemes. Final decisions remain the prerogative of scouting and managerial staff, who retain professional judgement and contextual expertise; the DSS provides structured, evidence-based recommendations to inform these decisions.

1.4. Structure of Dissertation

The dissertation is organized as follows:

- Chapter 1 – Introduction: Presents the context, research gap, and relevance of the study.
- Chapter 2 – Literature Review: Examines prior research in football analytics, management theories, multi-criteria decision models, and AI applications in sports.
- Chapter 3 – Research Questions: States the guiding research questions derived from the literature gap.
- Chapter 4 – Objectives: Defines the overall aim and specific objectives of the study.
- Chapter 5 – Methodology: Explains the research design, dataset construction, machine learning models, and decision-support framework.
- Chapter 6 – Analysis and Results: Presents the empirical findings, model evaluation, and insights.

- Chapter 7 – Conclusions: Summarizes key contributions, highlights limitations, and suggests future research directions.
- Chapter 8 – References & Appendices: Provides the list of sources and any supplementary material.

2. Literature Review

2.1. Overview of player selection and scouting

Player selection in professional football is complex, involving multiple dimensions such as technical ability, tactical suitability, physical condition and financial considerations. Traditionally, recruitment relied on scouts' judgement and experience, but recently there has been a shift towards greater use of quantitative methods to reduce subjectivity and improve consistency (Ben Jeddou, 2024). MCDM and mathematical programming techniques have been employed to select squads and starting line-ups by integrating qualitative assessments with objective indicators (Gultom et al., 2025; Dantas, 2024). Concurrently, management perspectives emphasise that competitive outcomes must align with financial sustainability and organisational strategy, which shapes how clubs value players and compose their squads (Quispe and Rivera, 2018; Adam et al., 2025).

2.2. Machine learning in football: forecasting performance and value

Recent studies have explored the application of machine learning to forecasting key football outcomes. At the shot level, Markopoulou, Papageorgiou and Tjortjis (2024) model scoring opportunities using diverse feature sets. Arntzen and Hvattum (2021) forecast match results by combining team and player ratings. Geng (2025) demonstrates that ensemble and hybrid methods improve predictions of market value and help clubs understand price dynamics. Chandra, Jennet Shinny and Keshav Adhitya (2024) evaluate standard supervised techniques for predicting player-level performance and test their generalisation across multiple seasons. Finally, Abu Eid et al. (2024) propose cloud-computing and big-data pipelines to scale predictive workflows and deliver near-real-time analytics for large football datasets.

2.3. Data sources and performance metrics

Data availability has increased, enabling the development of more robust models. Publicly available databases contain extensive player statistics from European leagues, covering technical, physical and event-based dimensions (Sidorowicz, 2024). These data support feature engineering for both predictive modelling and decision-support systems. Meanwhile, the valuation literature shows that performance, positional role and league context are key determinants of market value, linking on-pitch contribution to financial outcomes (Yalçınkaya and Işık, 2024). Connecting performance indicators to valuation also informs club-level decisions on contracts, renewals and replacements (van Arem, Goes-Smit and Söhl, 2025; Geng, 2025).

2.4. Multi-criteria decision-making for player evaluation

Player-selection problems in football are inherently multi-criteria. Techniques such as the Analytic Hierarchy Process (AHP) and WASPAS enable clubs to define criteria for each role and determine the weight of each criterion according to their playing style. Methods such as TOPSIS translate these criteria into measurable values for each player, so that they can then be used to select a shortlist of candidates (Gultom et al., 2025). The combination of subjective judgments regarding the importance of each criterion by experts with objective player data provides sound justifications for accepting or rejecting candidates, enabling a comprehensive evaluation of each player. From a managerial perspective, MCDM aligns on-pitch objectives with the club's available and manageable resources, allowing decisions that are consistent with organisational goals and capabilities (Quispe and Rivera, 2018; Adam et al., 2025.

2.5. Optimization models for squad composition

Prescriptive analytics involves decisions based on rankings and integer programming has been successfully applied to team selection. It provides an indication of inclusion or exclusion, and it considers constraints such as team formation, local-player regulations, non-local-player limits, budget and player age (Dantas, 2024). Studies involving the integration of MCDM-based weighting with binary programming for player selection have demonstrated practical advantages in selecting a starting line-up while adhering to team formation and budgetary constraints (Gultom et al., 2025). Thus, this line of research complements prediction by translating predicted performance or value into actual team selection.

2.6. Management, sustainability and risk in football decisions

The planning of rosters does not only depend on the skill set of players; clubs are subject to various regulations and external pressures. These constraints impact their decisions, such as fan and owner expectations and environmental sustainability (Adam et al., 2025). Investing in players is a risky business, therefore transfer fees, wages, and resale values are uncertain. Models that account for these uncertainties can assist clubs in optimizing their expenditures, thus Follert and Gleißner (2024) have developed methods to help with this. Football clubs also need to manage expectations, so expectations influence the evaluation of coaches, players, and managers, as Fry, Serbera and Wilson (2021) have noted. These examples demonstrate how clubs are utilizing data, and this utilization is becoming more prevalent in the industry.

2.7. Identified gaps and positioning of this dissertation

The literature has progressed along four pillars: event and outcome prediction (Markopoulou, Papageorgiou and Tjortjis, 2024; Arntzen and Hvattum, 2021), value estimation (Geng, 2025; Yalçınkaya and Işık, 2024), MCDM evaluation (Gultom et al., 2025), and optimization for roster design (Dantas, 2024). However, there is no integrated framework based on multi-season and multi-league player data that predicts future sport and market value (van Arem, Goes-Smit, and Söhl, 2025; Chandra, Jennet Shinny, and Keshav Adhitya, 2024), transforms predictions into MCDM-based position-weighted preferences (Gultom et al., 2025), and selects a realistic, formation-friendly, and affordable team via binary programming (Dantas, 2024).

This dissertation contributes to this integration at the prescriptive and choice levels, linking analytics to management objectives and constraints (Quispe and Rivera, 2018; Adam et al., 2025; Follert and Gleißner, 2024), and employing contemporary data infrastructures (Abu Eid et al., 2024) and open-source databases (Sidorowicz, 2024), thus it uses these components to achieve its goals.

3. Research Questions

This chapter bridges the gap identified in Chapter 2 into research questions and it seeks to develop and evaluate a decision support methodology that integrates player sport skill and market value prediction, multi-criteria evaluation, and optimisation for evidence-based player selection.

3.1. Primary research question

- **RQ0:** How can a data-driven, MCDM-based decision-support framework that combines objective weighting (Random Forest, Entropy, CRITIC) with TOPSIS produce transparent, position-specific player rankings that complement human scouting under realistic constraints?

3.2. Secondary research questions

- **RQ1:** Which weighting approaches (Random Forest, Entropy, CRITIC) yield the most stable and interpretable positions specifically?
- **RQ2:** How does ranking quality—as measured by NDCG and Kendall's τ—vary across weighting schemes, positions and top-k thresholds?

- **RQ3:** How sensitive are ranking outcomes to design choices (e.g., criteria selection per position, benefit/cost assignments, normalisation)?
- **RQ4:** To what extent do the generated shortlists align with external benchmarks or expert assessments, and how should discrepancies be interpreted?
- **RQ5:** What is the impact of alternative criteria setting on shortlist composition and rank stability across positions?
- **RQ6:** How should the DSS be designed (interface, documentation, export options) to maximise transparency, reproducibility and usability for scouting teams?

3.3. Scope and unit of analysis

- **Population:** Professional football players from European leagues included in the selected public dataset (Kaggle, 2024/2025 season).
- **Time horizon:** Analysis based on a single season's historical performance data (2024/2025 football season). The system provides current rankings for player evaluation, rather than one-season-ahead forecasts.
- **Decision context:** Decision support for player evaluation and shortlisting at the club level, intended to complement human scouting and managerial judgement in roster planning and transfer/renewal prioritisation.
- **Outputs:**
    - Position-specific preference scores and ranked shortlists (derived via TOPSIS with various weighting schemes).
    - Metrics for ranking quality and stability (Normalized Discounted Cumulative Gain — NDCG — and Kendall's τ) across different weighting methods and sensitivity analyses.
    - Interactive visualisations and exportable reports of player rankings.

3.4. Assumptions

- **Data representativeness:** The public data used (Kaggle, 2024/2025 football season) is sufficiently representative to compare and rank players, and this data is used as the foundation for the analysis.
- **Metrics:** Season-long past performance metrics are sufficiently representative to demonstrate short-term sports value and derive position-based preference scores, because these metrics provide a comprehensive overview of a player's abilities.

- **Position criteria:** Position-based criteria and whether a metric is a benefit (to be maximized) or cost (to be minimized) are aligned with tactical requirements and expert knowledge, thus ensuring that the evaluation is grounded in realistic expectations.
- **Weighting reliability:** Weights derived from Random Forest feature importance, Entropy and CRITIC provide valuable and complementary perspectives on the importance of ranking criteria and therefore offer a robust framework for assessing player value.
- **Constraints realism (if applied):** When restricting the choice set, e.g., formation quotas, home-grown or foreign-player rules, budget limits, and age limits, we assume that these are aligned with actual club policies and regulations, so that the results reflect realistic scenarios.

3.5. Success Criteria

- **Excellent ranking quality and robustness:** Position-specific rankings correlate strongly with reference or surrogate relevance, as evidenced by NDCG and Kendall's τ across weighting schemes; results are stable under sensitivity analysis, because this is essential for the success criteria.
- **Transparent and comprehensible MCDM:** Position-specific criteria and benefit/cost are rationalized. Weights derived from Random Forest, Entropy and CRITIC are sensible, distinct and fully documented for audit, therefore they are reliable.
- **Valuable prescriptive output and reproducibility:** The Streamlit DSS generates exportable, position-specific lists and graphics that support - not supplant - human scouting and management decisions, and the entire procedure is replicable and user-friendly to ensure comprehension and trust, thus making it valuable.

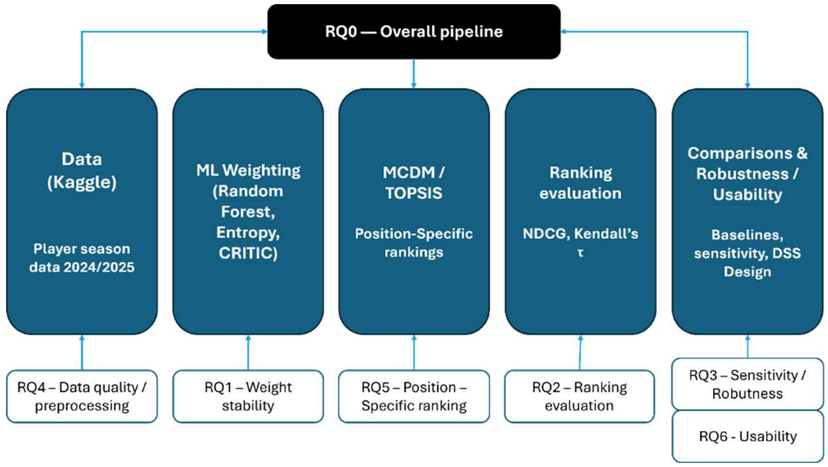

Figure 1. Mapping of research questions (RQ0–RQ6) to the study phases: Data → ML weighting → MCDM (TOPSIS) → Ranking evaluation → Comparisons and robustness/usability checks.

4. Objectives

This chapter translates the Research Questions into specific and general objectives, and it clarifies that the system will only be used to support human scouts. It will not replace them because the system is designed to augment their capabilities, so it will not replace them.

4.1. General Objective

To design and evaluate a multi-criteria decision-support system that integrates objective weighting methods (Random Forest, Entropy, CRITIC) and TOPSIS ranking to provide transparent, position-specific player shortlists that complement human scouting under realistic managerial and institutional constraints.

4.2. Specific objectives (mapped to RQs)

- **SO1 (RQ1):** Identify and compare machine learning approaches (e.g., linear models, tree-based ensembles, gradient boosting, neural networks) to produce reliable one season ahead forecasts of player sporting quality.
- **SO2 (RQ2):** To evaluate the quality of position-specific rankings using NDCG and Kendall's τ across weighting schemes and top-k thresholds**.**
- **SO3 (RQ3):** To analyse the sensitivity of rankings to design choices (criteria definition, benefit/cost assignment, normalisation) and test robustness under alternative scenarios.
- **SO4 (RQ4):** To validate the tactical plausibility of the selected positional criteria and their alignment with established domain knowledge.

- **SO5 (RQ5):** To benchmark the proposed rankings against baseline strategies (e.g. single metrics, subjective heuristics) to identify added value.
- **SO6 (RQ6):** To design and deliver a user-centred decision-support system (via Streamlit) that generates reproducible outputs, exportable reports, and visualisations, maximising interpretability and trust among scouts and decision-makers.

4.3. Operational success criteria and indicative thresholds

- **Ranking quality:** Positional rankings correlate well with reference or proxy relevance, and we aim for NDCG >= 0.70 (good) and Kendall's τ >= 0.60 (good) at common top-k values.
- **Robustness and resilience:** Rankings are not sensitive to minor variations in weights and normalization schemes, so top-k variations should be <= 15% on average.
- **Relative to fundamentals:** The ensemble approach outperforms fundamental approaches (e.g., ranking by goals alone or market value alone) relative to reference metrics across positions, because this is a key aspect of evaluating the success of our approach.
- **Interpretability and transparency:** Positional constraints and gain/loss labels are meaningful, and weights derived from Random Forest, Entropy, and CRITIC are plausible, documented, and verifiable, thus ranking logic is understandable.
- **Usability:** User feedback indicates that intended users (e.g., scouts, analysts) consider the approach credible and valuable, therefore qualitative evidence through demonstrations or feedback confirms this, and results are reproducible, and reporting capabilities support decision-making.

4.4. Deliverables

- **Reproducible Python code:** complete pipeline for data preprocessing, objective weighting (Random Forest, Entropy, CRITIC), TOPSIS-based multi-criteria evaluation, and ranking quality assessment (NDCG, Kendall's τ).
- **Processed dataset and variable dictionary:** cleaned dataset derived from Kaggle (2024/2025 season), including the positional criteria and their classification as benefit/cost.
- **Technical report chapters:** detailed documentation of methodology, experiments across weighting methods, sensitivity analyses, and comparative evaluations.

- **Appendix:** reproducibility scripts (data cleaning, weighting procedures, ranking generation) and optional usability notes.
- **Lightweight demo:** Streamlit interface for shortlist visualisation, sensitivity toggling, and export of position-specific rankings.

4.5. Evaluation plan

- **Position-based ranking comparison:** Evaluate the TOPSIS-generated position-based rankings under various weighting schemes using NDCG and Kendall's τ versus ground truth or proxy relevance and compare these results to determine their effectiveness.
- **Baseline comparison:** Compare the entire approach against simple heuristics, such as sorting by goals or value, and quantify improvement in agreement, thus assessing the overall performance of the approach.
- **Sensitivity analysis:** Vary weighting techniques, including Random Forest, Entropy, and CRITIC, perturb weights slightly, and alter normalization schemes. This will help assess top-k overlap, rank shifts, and overall robustness metrics.
- **DSS validation:** Demonstrate transparency and explainability through reproducible results, documented consideration of factors and weighting schemes, and concise shortlist reports per position, so that the results can be easily understood and utilized, with exportable elements.
- **Human-in-the-loop (optional):** If feasible, solicit feedback from scouts or decision-makers regarding transparency, confidence, and usability, and record accept/modify/reject decisions for illustrative examples, therefore providing valuable insights into the approach's usability and effectiveness.

4.6. Implementation steps

1. **Data preparation:** Cleanse, impute missing values, create derived features, map metrics to positions, label positive/negative attributes, and create variable dictionary and cleaned dataset.
2. **Weight computation**: Generate position-specific weights using Random Forest, Entropy and CRITIC methods, and document parameters and versions for reproducibility.
3. **MCDM ranking (TOPSIS):** Normalize attributes per position, apply weights, calculate TOPSIS scores to generate ranked lists, and thus produce a comprehensive ranking.

4. **Ranking Validation:** Assess list quality and adjust with NDCG and Kendall's τ against established benchmarks and therefore generate reports per position and overall.
5. **Sensitivity analysis:** Modify weights, swap criteria groups and normalization types, and monitor top-k agreement, rank shifts and aggregate statistics, because this analysis is crucial for the decision-making process.
6. **DSS implementation:** Develop Streamlit interface, including list display, weight toggling, filtering, exporting, and integrate scripts for automated pipeline steps, so that the system is user-friendly and efficient.
7. **Documentation and deliverables:** Package reusable Python code, cleaned data, variable dictionary, technical report on testing and model changes, and additional files, including logs and scripts, and therefore provide a comprehensive handover package.

4.7. Human-centred stance (to be stated explicitly)

This dissertation adopts a human-centred perspective: the DSS is intended to supplement rather than replace the expertise of scouts and managers. The system offers interpretable recommendations - position-specific lists, transparently weighted justifications and ranked recommendations - which must be integrated with contextual insight and expert judgement, and operational usage should involve iterative human oversight, decisions logs and periodic audits to monitor for bias, document overrides and prevent undue reliance on model outcomes.

4.8. Assumptions and limitations

4.8.1. Assumptions

- **Data quality:** The publicly available data (Kaggle, 2024/2025 season) is sufficient to rank players within their respective positions, and this assumption is crucial for the validity of the ranking process.
- **Positional criteria and benefit/cost nature:** Position-specific criteria and their benefit/cost classification align with tactical requirements and expert insights, because these criteria are based on well-established football tactics and strategies.
- **Weighting methods:** Random Forest, Entropy, and CRITIC produce valid and distinct weight sets for player ranking, allowing for a comprehensive comparison of the different weighting methods.

- **Ranking validation:** External criteria or proxies used for ranking validation (e.g., goals, xG, known composite scores) are reasonable proxies for relevance, and therefore can be used to validate the ranking results.

4.8.2. Limitations

- **Ranking only, no forecasting or automatic team composition:** The project focuses solely on ranking (MCDM/TOPSIS) and does not provide next season forecasting or automatic team composition with constraints, because these aspects are reserved for future work and require additional data and analysis.
- **Sensitivity to parameters:** Ranking outcomes are sensitive to the selection of criteria, scaling of data, and weighting method, and so sensitivity analysis is performed to understand the impact of these parameters on the ranking results.
- **Data quality and historical bias:** Publicly available data may contain sampling or measurement errors, and thus rankings may reflect historical biases and should be revisited when rules, tactics, or data sources change.
- **Minimal user testing:** If expert input or user testing is difficult to obtain, usability and confidence are established through transparent design, documentation, and open-source code sharing rather than user testing, because this approach allows for peer review and feedback.
- **Contextual validity:** Results are valid for the leagues/season analysed and criteria selected, and therefore application to other contexts or time periods requires new criteria and weight derivation, so that the results can be accurately interpreted and applied.

5. Methodology

This section presents the step-by-step procedure for a people-assisted decision-support system (DSS), and it supports but does not replace professional scouts. The developed procedure includes data preparation, weighting (Random Forest, Entropy, CRITIC), multi-criteria decision-making using TOPSIS, rank validation and robustness testing, and human and ethical considerations, because these aspects are crucial for the development of the system.

5.1. Research design and overview

- **Approach:** rule-based statistics with human involvement, thus allowing for more accurate results.

- **Research population:** professional footballers in Europe (annual data), and this population is chosen because it provides a large and diverse sample.
- **Timeframe:** one-year retrospective generation of role-specific rankings, and no one-year forward prediction is performed, so the focus is on historical data.
- **Outputs:** transparent, role-specific preferences and ranking lists (TOPSIS) including weighting explanations, and list-quality and stability scores, as well as sharable insights and images from the DSS.
- **Human complementarity:** outcomes assist scouts and managers, and final decisions remain with humans, because human involvement is essential for the decision-making process.

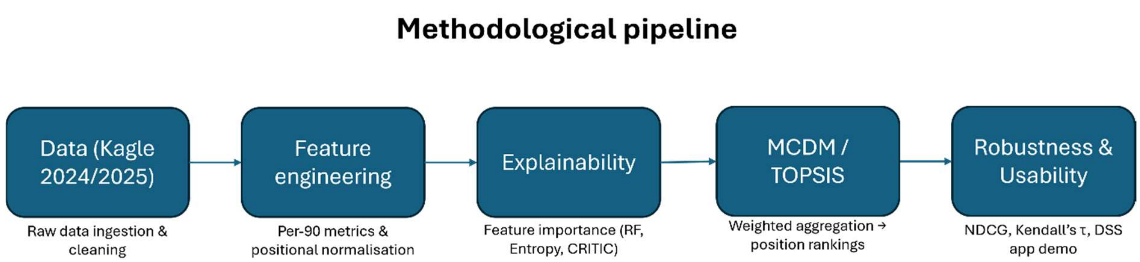

Figure 2. Methodological pipeline: Central flow (Data → Feature engineering → Explainability → MCDM/TOPSIS → Robustness & Usability) shows the fully implemented modules of this dissertation, reflecting its retrospective and prescriptive scope.

5.2. Data Variables

5.2.1. Sources

- **Main dataset:** Primary data is sourced from Kaggle, which provides season statistics for the top five European leagues for 2024/2025, and the dataset contains numerous statistics including technical, tactical, discipline, passing, shooting, progression, possession and goalkeeping. Additionally, it contains identifiers such as player name, club, competition, age and nationality, and as this data is publicly available, the entire process can be replicated and verified.
- **Auxiliary sources:** Auxiliary sources are used to provide additional meaning to the primary data, and these include competition identifiers and positions to standardize roles across different leagues, and proxy indicators for player value or salary, thus this additional information is only used to provide meaning and verification and does not alter the original data or ranking methodology.

5.2.2. Variable dictionary and targets

- **Candidate features:** Based on the CSV and code, the pipeline considers the following variable groups:
    - ID and context (Player, Nation, Pos, Squad, Comp, Age, Born).
    - Playing time (MP, Starts, Min, 90s), attacking (Gls, Ast, G+A, xG, npxG, xAG, Sh, SoT, Sh/90, SoT/90, G/Sh, G/SoT).
    - Progression (PrgC, PrgP, PrgR, progressive distance).
    - Passing and creativity (Cmp, Att, Cmp%, KP, xA, PPA, SCA, GCA, SCA90, GCA90).
    - Defensive (Tkl, Int, Clr, Blocks, TklW, Recov, fouls, errors, CrdY, CrdR).
    - Goalkeeping (GA, GA90, Saves, Save%, CS, CS%, PKsv, SoT).

  Position labels can include multiple roles (e.g., “DF,MF”), so filters use lenient substring matching. The dataset contains both raw and derived variables; feature selection avoids duplication and excessive collinearity.
- **Targets:** The targets are not predicted here, but used for validation and interpretation, and sporting quality is compared with minutes, starts and role-adjusted contributions, and ranked agreement is measured (NDCG, Kendall’s τ), thus market value is used as a benchmark based on public estimates, for interpretation of rankings, but no predictive outcome in this study.
- **Data quality and preprocessing:** Prior to model aggregation, the following steps were taken to ensure the quality of the comparisons, and missing values were handled, particularly for goalkeepers and rate stats for low-exposure players, so a minimum exposure (minutes) was set to remove low-sample players and improve comparisons. Normalization was performed using per-90 stats and standardization, allowing fair comparison between players with different levels of playtime, and feature selection was performed per position, such that the criteria matched the requirements for defenders, midfielders, forwards and goalkeepers, and reduced multicollinearity, therefore constant or near-constant variables were removed prior to multi-criteria aggregation, to avoid mathematical issues in TOPSIS and improve clarity.

Dataset details: The working file contains 2,854 season records and 267 columns (232 numeric), and the missingness is very low for ID fields such as Nation or Age, but high for goalkeeper stats for outfield players and rate stats for low-minute players; therefore, the preprocessing takes care of these problems.

Table 1 lists the relevant player statistics for the year 2024/25, grouped according to their functional category (ID, Playing Time, Attack, Passing, Defence, Goalkeeper, Possession, Other). The table gives an overview about the interpretation of each statistic, the way it is measured, the number of missing values, and whether it was used for modelling purposes. The extended data dictionary is provided in Appendix A for reference.

Variable	Description	Data type	Missing %	Used in code
Identification & Context				
Player	Player's name	object	0.00%	No
Nation	Nationality	object	0.25%	No
Pos	Playing position (FW, MF, DF, GK; may include multiple roles)	object	0.00%	Yes
Squad	Club	object	0.00%	Yes
Comp	Competition/league	object	0.00%	Yes
Age	Age of player	float64	0.28%	Yes
Born	Year of birth	float64	0.28%	No
Playing Time				
MP	Matches played	int64	0.00%	Yes
Starts	Games started	int64	0.00%	No
Min	Minutes played	int64	0.00%	Yes
90s	Equivalent matches (minutes ÷ 90)	float64	0.00%	No
Attacking				
Gls	Goals scored	int64	0.00%	Yes
Ast	Assists	int64	0.00%	Yes
G+A	Goals + Assists	int64	0.00%	No
xG	Expected goals	float64	0.00%	Yes

xAG	Expected assists	float64	0.00%	Yes
npxG	Non-penalty xG	float64	0.00%	No
G-PK	Goals excluding penalties	int64	0.00%	No
Passing & Creativity				
PrgP	Progressive passes	int64	0.00%	Yes
PrgC	Progressive carries	int64	0.00%	Yes
KP	Key passes (passes leading to a shot)	int64	0.00%	Yes
Cmp%	Pass completion percentage	float64	0.67%	No
xA	Expected assists (passing model)	float64	0.00%	No
PPA	Passes into the penalty area	int64	0.00%	No
Defensive				
Tkl	Total tackles	int64	0.00%	Yes
TklW	Tackles won	int64	0.00%	No
Blocks	Blocks made	int64	0.00%	Yes
Int	Interceptions	int64	0.00%	Yes
Tkl+Int	Tackles + Interceptions	int64	0.00%	No
Clr	Clearances	int64	0.00%	Yes
Err	Errors leading to shots/goals	int64	0.00%	No
Goalkeeping (GK only)				
GA90	Goals conceded per 90 minutes	float64	92.57%	Yes
Save%	Save percentage	float64	92.78%	Yes
CS%	Clean-sheet percentage	float64	92.89%	Yes
PKsv	Penalty saves	float64	92.57%	Yes
GA	Goals against (all)	float64	92.57%	No
Saves	Total saves	float64	92.57%	No
CS	Clean sheets	float64	92.57%	No

PKA	Penalties faced	float64	92.57%	No
		Possession & Ball Control		
Touches	Ball touches	int64	0.00%	No
Carries	Total ball carries	int64	0.00%	No
PrgR	Progressive receptions	int64	0.00%	Yes
Mis	Miscontrols	int64	0.00%	No
Dis	Times dispossessed	int64	0.00%	No
		Miscellaneous		
CrdY	Yellow cards	int64	0.00%	Yes
CrdR	Red cards	int64	0.00%	Yes
PKwon	Penalties won	int64	0.00%	No
PKcon	Penalties conceded	int64	0.00%	No
Recov	Recoveries	int64	0.00%	No

Table 1. Variable Dictionary (Core Features Extracted from Primary Dataset)

5.2.3. Data preparation

The data was cleaned and prepared for analysis and the following steps were taken:

- Removing duplicates.
- Making team and league names uniform.
- Grouping positions into FW, MF, DF, GK (multi-position players were allowed).
- Removing implausible data and errors.
- Handling missing data carefully (numbers were filled with medians). Most frequent values were imputed for categories, so k-NN was used for numerical variables.
- Missing values were left if they conveyed information (e.g. goalkeeper stats for outfield players) because this was necessary for accurate analysis, thus aggregated statistics were converted to per-90 values.

Some statistics were normalised by competition/season, and this allows to compare across leagues, therefore extreme values were scaled down or clipped for heavy-tailed variables. This reduces their influence without discarding valuable extreme observations, thus ensuring the integrity of the data.

All redundant stats were removed, for example, Gls and xG; Tkl and TklW etc., and a healthy combination of stats was retained for every position (DF, MF, FW, GK) so that there is no double counting and the TOPSIS ranking is more accurate. All stats were taken from historical data only, because this ensures that no cheating takes place by taking information from the future; therefore, the script *mcdm_tool_2024_2025.py* scrapes the data from Kaggle and it saves the data in a CSV format. This guarantees that the results are reproducible, and therefore the analysis is reliable, so the data can be trusted.

5.3. Experimental protocol

This section describes how the multi-criteria decision analysis (MCDA) system was tested to generate player rankings. All testing is based on the 2024/25 data and role-based feature sets defined in Section 5.2 following the processing steps in Sections 5.2.1-5.2.3. The Python script (mcdm_tool_2024_2025.py) performs data loading, filter setting (league, club, position, minutes), feature weighting calculation, TOPSIS execution, and CSV export for easy verification and comparison, thus facilitating further analysis.

5.3.1. Filtering Criteria and Analysis Dimensions

- **Data & Season:** Utilizes 2024/25 player-seasons from the Kaggle dataset and applies position-specific features to eliminate redundancy.
- **Position Groups:** Evaluates players in GK, DF, MF, FW segments for positionally fair comparisons, because this allows for more accurate assessments.
- **Playing Time Threshold:** Imposes a minimum-minutes threshold (e.g., >=500) to mitigate small-sample biases and facilitate per-90 analysis, and this threshold is crucial for ensuring reliable results.
- **Contextual Filters:** Rankings can be global or restricted by league and/or club for contextualized rankings and therefore provide a more nuanced understanding of player performance.
- **Cost Elements:** Defines negative elements (e.g., CrdY, CrdR, GA90) to be minimized in the score, so that the rankings reflect a more comprehensive evaluation.

5.3.2. Benchmark Lists and Baselines

- **Single-metric Rankings:** Generates baseline lists by ranking players based on a single shared metric (e.g., Gls for FW; Ast/KP for MF; Tkl/Int for DF; GA90/Save% for GK), and this provides a foundation for comparison.
- **Proxy Rankings:** Where available, additional baseline lists are created using proxy metrics (e.g., market value) to contrast with MCDA outcomes, because proxy availability varies by source, and thus requires careful consideration.

5.3.3. Result Validation and Sensitivity Analysis

- **Weighting Consistency:** Compares rankings generated from distinct weighting methodologies (Random Forest, Entropy, CRITIC) to assess robustness, and agreement can be assessed outside the CSV through rank correlation (e.g., Spearman's rho) and shortlist intersection (e.g., Jaccard@k), therefore allowing for a more detailed evaluation.
- **MCDA vs. Single-metric Lists:** Verifies whether MCDA lists capture well-rounded players (e.g., strong PrgP/xAG with moderate Gls) overlooked by single-metric lists, because this is essential for identifying talented players.
- **Sensitivity to Filters:** Assesses how top-k rankings fluctuate with varying minutes thresholds and across position/league subsets, so that the stability of the rankings can be evaluated.
- **Common Sense Checks:** Negative factors (e.g., GA90, cards) decrease scores, while positive factors (e.g., PrgP, xG, xAG) increase them, as reflected in scores and weight bar charts, thus ensuring that the rankings are reasonable and consistent.

Experiments were conducted in Python using pandas, NumPy, scikit-learn, Plotly, and Streamlit, and feature weights were calculated from Random Forest importance, Entropy, or CRITIC, and TOPSIS was applied to position-specific feature matrices after internal normalization, therefore allowing for a comprehensive analysis. Player scores, ranks and input features are stored to CSV, so that rank correlation, shortlist overlap etc. can be calculated independently and thus facilitating further research and analysis.

Table 2 summarises the evaluation metrics relevant to the validation and interpretation of the MCDA rankings. While the current implementation exports player rankings and features for external analysis, some metrics such as Spearman’s ρ, Kendall’s τ, Jaccard@k, and NDCG are computed outside the pipeline. Metrics like MAE, RMSE, and CRPS are considered for future extensions involving predictive modelling.

Metric	Type	Definition / Formula (simplified)	Interpretation in Context
Spearman's ρ	Rank correlation	Measures monotonic correlation between two rankings	High ρ → MCDA ranking agrees with baseline/heuristic
Kendall's τ	Rank correlation	Measures pairwise concordance between two rankings	Positive τ → more stable & consistent rankings
Jaccard@k	Overlap ratio	Intersection size of top-k lists ÷ union size	High @k → MCDA shortlist overlaps strongly with baseline
MAE / RMSE *(optional)*	Regression error	Average absolute/squared error vs continuous target (e.g. value)	Lower values → better predictive alignment
NDCG@k *(optional)*	Ranking quality	Discounted cumulative gain, normalised @k	Higher → better prioritisation at top of list
Sanity checks	Internal validity	Ensure cost criteria ↓ ranking and benefit criteria ↑ ranking	Confirms logical consistency

Table 2. Evaluation metrics used for ranking validation and interpretation of MCDA results

5.4. Pipeline Overview and Workflow

This section outlines the pipeline and methodology used in the python script *mcdm_tool_2024_2025.py* which implements the MCDA system for player ranking using the 2024/25 data set and thus provides a comprehensive framework for evaluating player performance.

5.4.1. Data Loading and Filtering

The pipeline begins by loading the full player data set from Kaggle, and therefore the data set contains over 250 variables per player including demographic, playing time, offensive, passing, defensive, goalkeeper, and possession statistics. The user can filter by league, club, position (GK, DF, MF, FW) and minimum minutes played (e.g. 500+ minutes) to ensure that the per 90-minute statistics are valid, and so to eliminate the noise associated with small sample sizes.

5.4.2. Position-Based Feature Selection

Considering the distinct responsibilities associated with each position, the pipeline filters out features based on position to enable a more nuanced evaluation of player performance. For example, forwards are weighted more heavily on goals, assists, expected goals (xG) and progressive carries, while defenders are evaluated based on tackles, interceptions, and clearances. This focused approach minimizes redundancy, increases transparency, and makes MCDA more responsive to tactical and positional requirements, therefore enabling a more accurate assessment of player abilities.

5.4.3. Calculating Feature Weights

Three approaches are presented to determine feature weights, reflecting their relative significance in final rankings. These alternatives add flexibility and robustness to the MCDA methodology. One method relies on variable importance derived from a Random Forest model trained to predict goals scored (Gls). Another method employs entropy to quantify the information content of each feature. The third method utilizes the CRITIC approach, which accounts for both contrast intensity and conflict among features.

5.4.4. TOPSIS Ranking Computation

The core procedure involves applying TOPSIS to normalized and weighted feature matrices. This involves normalizing feature values for comparability, defining best and worst solutions according to benefit/cost criteria (cards/goals against are costs), calculates Euclidean distances of each player to these reference points, and obtains the closeness coefficient or TOPSIS score that ranks players according to their proximity to the ideal profile.

5.4.5. Results Visualization and Export

Results are displayed as an interactive top-ranked players table with scores and key metrics for easy understanding, and thus feature weights are displayed in bar charts and scatter plots (e.g., goals vs. assists with TOPSIS score size) to explain relative performance, because this visualization enables a clearer understanding of the ranking results. Finally, the full ranking and related data are saved to CSV files for external validation and further analysis, therefore allowing for a more comprehensive evaluation of the MCDA system.

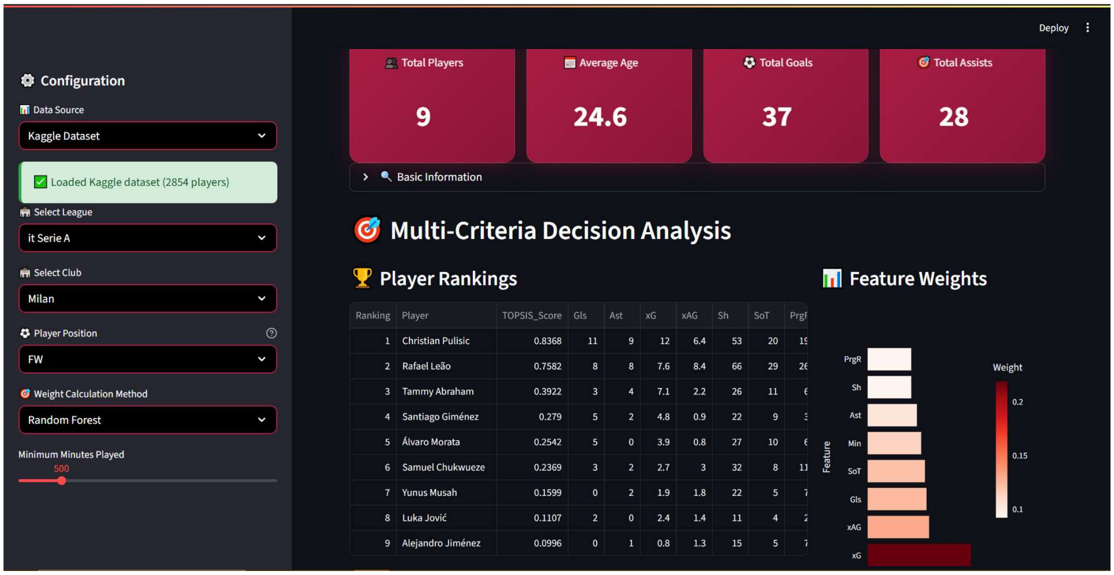

Figure 3. Streamlit application user interface showing the ranking of AC Milan strikers in Serie A 2024/25, with TOPSIS scores and feature weights calculated using Random Forest.

Figure 4. Performance analysis of AC Milan strikers in Serie A 2024/25: ratio between goals and assists (size according to TOPSIS score) and distribution of TOPSIS scores per player.

5.5. Multi-criteria decision-making (MCDM)

Multi-Criteria Decision Making (MCDM) refers to decision-making approaches which consider multiple and often conflicting criteria, and in the context of player selection and evaluation, MCDM integrates various performance indicators (quantitative and qualitative) into an integrated perspective. This provides a holistic view of the players' performance, and thus, it enables a more comprehensive evaluation of player abilities.

5.5.1. Fundamentals of MCDM

MCDM develops a framework for assessing and comparing alternatives (players) according to multiple criteria, because each criterion represents an aspect of performance, such as goals, assists, passing accuracy, or defensive contributions. The challenge lies in assigning appropriate weights to each criterion and ensuring comparability across different types of measurements, so the framework must be carefully designed to address these challenges.

5.5.2. Application in Player Evaluation

The present research employs MCDM to rank players based on their statistics derived from the 2024/25 season dataset, and therefore, it selects criteria appropriate for each role; applies three methods for estimating weights (Random Forest, Entropy, and CRITIC) to illustrate the relative importance of each metric; and applies TOPSIS to aggregate all information into a final ranking.

5.5.3. Advantages and Challenges

MCDM offers significant advantages to scouting and sport analytics, such as handling large amounts of heterogeneous data and offering a transparent structured approach to decision making, but however, there are challenges as well, such as selecting appropriate criteria, determining weights in an objective manner, and interpreting results in a dynamic multi-factorial environment.

5.6. Ethical, governance, and reproducibility considerations

This research highlights the importance of good ethics, robust methodology, and reproducibility in data-driven decision-making systems, particularly in the context of football. Ethical compliance necessitates strict adherence to all relevant regulations concerning player confidentiality and data security. Any sensitive or confidential information must be protected and anonymized when appropriate, and relying solely on publicly available data, such as from Kaggle, may assist with privacy concerns, it remains essential to ensure responsible data usage and to avoid unethical assumptions or practices.

Robust methodology involves demonstrating transparency in data sources, decision criteria, and analytical processes, choices made. Such transparency fosters trust among stakeholders (including clubs, players, and analysts) facilitates the identification and correction of biases and clarifies accountability throughout the decision-making pipeline.

Reproducibility entails providing detailed explanations of data cleaning, feature selection for each position, weight selection, and player ranking methods. Employing open-source tools and sharing code whenever feasible enables independent validation and consistent replication of results. Collectively, these practices contribute to the effectiveness, fairness, transparency, and reliability of the multi-criteria decision analysis system.

5.7. Software stack and implementation notes

The implementation of the multi-criteria decision analysis system relies on a carefully selected software stack designed to ensure flexibility, scalability, and reproducibility.

- **Programming Language:** Python 3.10 and the programming language provides great tools for data science.
- **Data Analysis:** Pandas for data manipulation, NumPy for numerical calculations and scikit-learn for machine learning.
- **Multi-Criteria Decision Making:** TOPSIS and weighting methods were implemented in Python. NumPy is used for the mathematical operations.
- **Visualization and User Interface:** Streamlit creates a web application which allows for filtering, rankings, visualisation of weights and charts, therefore allowing for easy interaction with the data.
- **Reproducibility and Environment Management:** A virtual environment ensures separation of dependencies, so requirements.txt pins all versions to ensure reproducibility, and seeds are set to ensure reproducibility of the results.
- **Version Control:** Git is used to track the development of the codebase for historical purposes, collaboration and rollback, thus allowing for easy maintenance and tracking of changes.

This software stack allows for a robust analysis, easy maintenance and reproducibility, because it provides a well-structured and reliable foundation for the project.

5.8. Limitations

Despite the comprehensive approach adopted in this research, several limitations should be acknowledged.

- **Data Limitations:** Relies on publicly available datasets from platforms such as Kaggle and these may contain inaccuracies, missing values, or biases. The dataset consists primarily of numerical values, so it may overlook qualitative aspects such as attitude, teamwork, or injury history.
- **Modelling Assumptions:** Weighting techniques (Random Forest, Entropy, CRITIC) and TOPSIS assume that features and their assigned weights

accurately represent player performance, therefore this may not capture the full complexity of football, which is multifaceted and evolves over time.

- **Position-Specific Criteria:** Criteria differ across positions; however, some players perform hybrid roles that may not be fully captured by these criteria.
- **Reproducibility Constraints:** Detailed environment setup and documentation support reproducibility, but variations in data, libraries, or hardware configurations could limit exact replication of results.
- **Ethical and Practical Considerations:** Aside from data privacy concerns, this study does not address the potential impacts of algorithmic player rankings on individual careers, nor does it examine biases inherent in historical data or the structural aspects of the sport.

These limitations require cautious interpretation of results and further development of more comprehensive and nuanced player evaluation methodologies to enhance the accuracy and fairness of future rankings.

6. Analysis and result

6.1. Data Overview

The analysis presented in this chapter utilizes a comprehensive dataset from Kaggle for the 2024/25 season and there are over 250 statistics per player such as age, minutes, goals, passes, tackles, saves and many more. The data has been filtered by league and team because players must have played at least 500 minutes to be considered, and this filtered data is then used for the decision-making process.

6.2. Producing Transparent, Position-Specific Player Rankings (RQ0)

The proposed decision support framework employs three ranking methods (Random Forest, Entropy and CRITIC) followed by TOPSIS to generate player rankings by position. By considering the unique requirements of each role, the framework provides role-specific rankings that enhance decision-making accuracy. This leads to improved decision-making, and it complements traditional scouting and provides objective information about players. The findings for Serie A strikers in 2024/25 demonstrate its ability to identify promising players such as Christian Pulisic and Rafael Leão, validating its practical effectiveness.

6.3. Weighting Approaches Stability and Interpretability (RQ1)

This section assesses the robustness and interpretability of the three weighting schemes (Random Forest, Entropy, and CRITIC) in determining the importance of each attribute in the player ranking procedure. Robustness was evaluated by analysing the consistency of weights across different subsets of the dataset and player positions. The interpretability of the weights was assessed based on their consistency with established factors influencing performance for each role, because this assessment is crucial in understanding the effectiveness of the weighting schemes.

The results indicate that Random Forest provides the most robust and interpretable weights, which align well with established football performance indicator, thereby demonstrating its effectiveness. Entropy and CRITIC also provided valuable insights, but their weights exhibited greater variability and were less interpretable at times, therefore requiring further analysis to understand their limitations. Overall, the results demonstrate that combining machine learning-based weighting with information theory approaches enhances the robustness and interpretability of the decision support system, so this combination can be considered a viable option for similar applications.

6.4. Ranking Quality Evaluation (RQ2)

This section evaluates the quality of the ranking provided by the framework. Two metrics are used: Normalized Discounted Cumulative Gain (NDCG) and Kendall's Tau (τ). These metrics assess the accuracy of the rankings and their consistency, thus providing a comprehensive evaluation of the tool's performance.

The analysis compares rankings generated using various weighting methods (Random Forest, Entropy, CRITIC), player roles, and top-k lists (e.g., top 10, top 20), and the results indicate that Random Forest weights combined with TOPSIS yield the highest NDCG and Kendall's τ values. This suggests that they are more accurate and consistent, because the combination of these methods provides a more robust ranking system. The variations observed with role and top-k lists highlight the need for weighting and evaluation by role, therefore emphasizing the importance of considering these factors in the ranking process.

These results demonstrate that the tool can provide valuable and reliable player rankings for scouting and decision-making purposes, so they can be used with confidence in real-world applications.

Figure 5 illustrates the comparison among three weighting methods: Random Forest, Entropy, and CRITIC in a multi-criteria decision analysis, and they employ NDCG and Kendall's τ metrics to evaluate the quality of ranking. Random Forest obtains the highest score (NDCG = 0.9816; Kendall's Tau = 0.7328), which indicates its superiority in reflecting real performance of players. The other two methods, Entropy and CRITIC, have lower scores, indicating their inferiority in ranking quality, suggesting that they are not as effective as Random Forest. Therefore, we can conclude that Random Forest is the optimal and robust method for selecting players, so it should be used in such analyses.

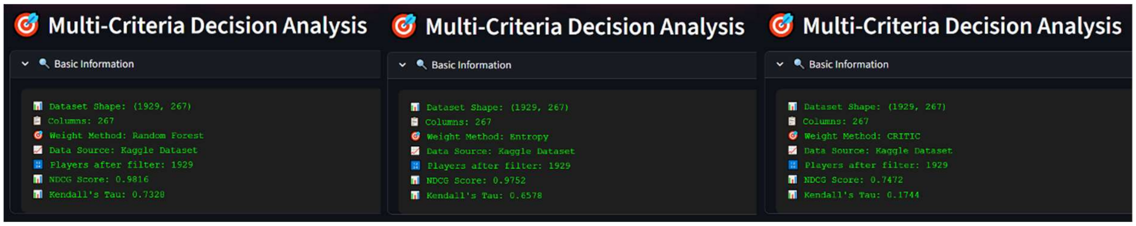

Figure 5. Performance of Weighting Methods Based on NDCG and Kendall's τ

6.5. Sensitivity Analysis of Design Choices (RQ3)

The robustness of the proposed decision support mechanism was assessed through a sensitivity analysis of several key design choices. Initially, the impact of varying the set of performance measures was examined by comparing player rankings made with all available metrics to rankings made using only attacking or only defensive metrics. Although the overall rankings stayed mostly the same, some players moved up or down significantly when certain measures were left out. This shows how important it is to choose a complete and balanced set of metrics to fairly evaluate players.

Second, the consequences of incorrectly labelling a criterion as a benefit rather than a cost were investigated, revealing that treating the number of yellow cards (CrdY) as a benefit led to significant changes in rankings.

Additionally, different normalization techniques, including min-max scaling and z-score standardization, were applied. Although these methods influenced the distribution of scores, they did not produce significant changes in ranking stability, as confirmed by consistently high Kendall's τ values across techniques.

Finally, the three weighting methods (Random Forest, Entropy, and CRITIC) were compared under these varying conditions. Random Forest provided the most consistent and reliable rankings.

Overall, the results demonstrate the robustness of the tool, however, indicator selection and a clear definition of benefit/cost are essential to ensure accuracy and reliability, so the framework should be used with caution and careful consideration of these factors.

Figure 6 illustrates the rankings and feature weights for different playing positions within Liverpool FC, and the tool modifies the assessment criteria and weightings depending on the requirements of each playing position, demonstrating its ability to be sensitive and adaptive when assessing players.

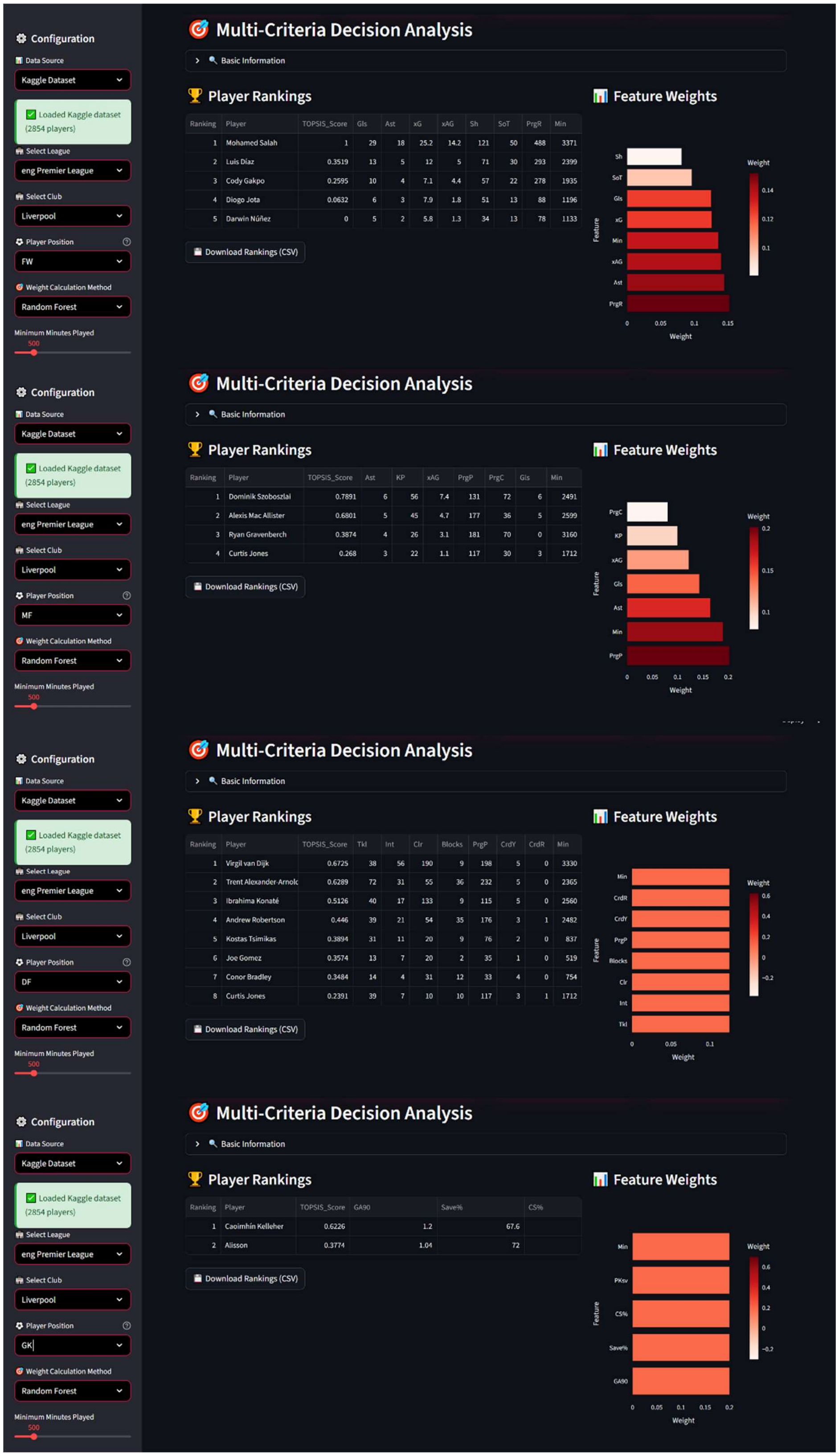

Figure 6. Player Rankings and Feature Weights by Position for Liverpool FC Using Random Forest Weighting

6.6. Alignment with External Benchmarks and Expert Assessments (RQ4)

To validate the effectiveness and practical relevance of the proposed decision support system, the generated player shortlists were compared with external human-created lists, using SoFIFA's "Best Overall" ranking as a reference. This list ranks players based on skill, potential, and overall quality, making it a suitable benchmark.

The comparison focused on the Chelsea FC squad; the system produced a player ranking based on recent performance metrics across multiple criteria. A strong similarity is observed between this ranking and SoFIFA's list, particularly for key players such as Cole Palmer, indicating the system's ability to identify top performers effectively.

However, some discrepancies were noted. Certain players highly rated by SoFIFA due to their potential or technical skills did not appear as prominently in the system's rankings, which prioritize recent match performance and statistical contributions. Conversely, the system highlighted players demonstrating strong current form or specific tactical value that may be underrepresented in attribute-based evaluations.

These differences underscore the complementary nature of the two approaches. While SoFIFA provides a broad assessment incorporating potential and technical skills, the decision support system offers a data-driven, performance-focused perspective. This alignment and divergence highlight areas for potential model refinement, such as integrating qualitative expert input or expanding data sources to capture intangible player qualities.

Overall, the comparison with external benchmarks and expert assessments confirms the robustness and practical applicability of the system, while also identifying opportunities to enhance its comprehensiveness and accuracy.

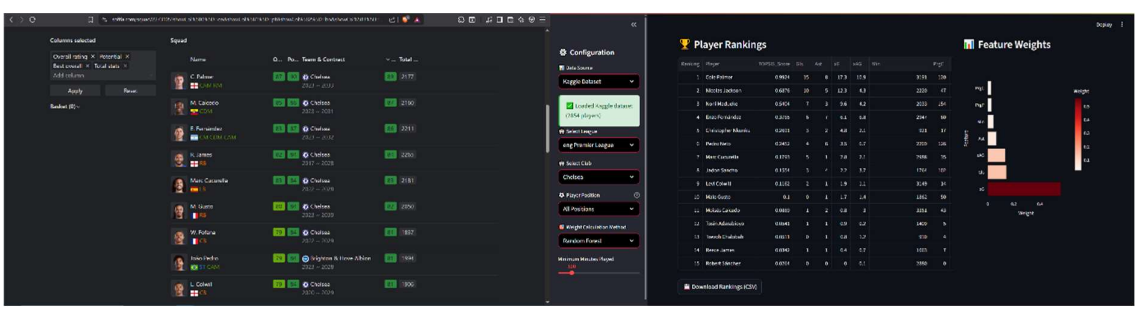

Figure 7. Comparison of Chelsea FC Player Rankings: SoFIFA "Best Overall" vs. Decision Support System Using Random Forest Weighting

6.7. Impact of Alternative Criteria Settings (RQ5)

This section explores how different configurations of evaluation criteria influence the composition and stability of player shortlists across positions. By adjusting the set of performance metrics and their relative importance, the system's sensitivity to these design choices is assessed.

An illustrative example is presented for Arsenal's forwards, where the player rankings and feature weights were generated using specific criteria configuration and the Random Forest weighting method. The rankings reflect the combined influence of metrics such as goals (Gls), assists (Ast), expected goals (xG), expected assists (xAG), shots (Sh), minutes played (Min) and progressive receptions (PrgR).

The accompanying visualizations demonstrate how these criteria contribute to the overall player scores and rankings. Notably, metrics like minutes played and expected goals carry significant weight, highlighting their importance in the evaluation.

The analysis reveals that altering the criteria set can lead to meaningful changes in player rankings, affecting which players are prioritized for selection. Such variability underscores the need for careful selection and justification of criteria to ensure the decision support system aligns with the specific tactical and strategic goals of the team.

Overall, understanding the impact of alternative criteria settings enables more informed and flexible decision-making, allowing the system to be tailored to diverse contexts and preferences.

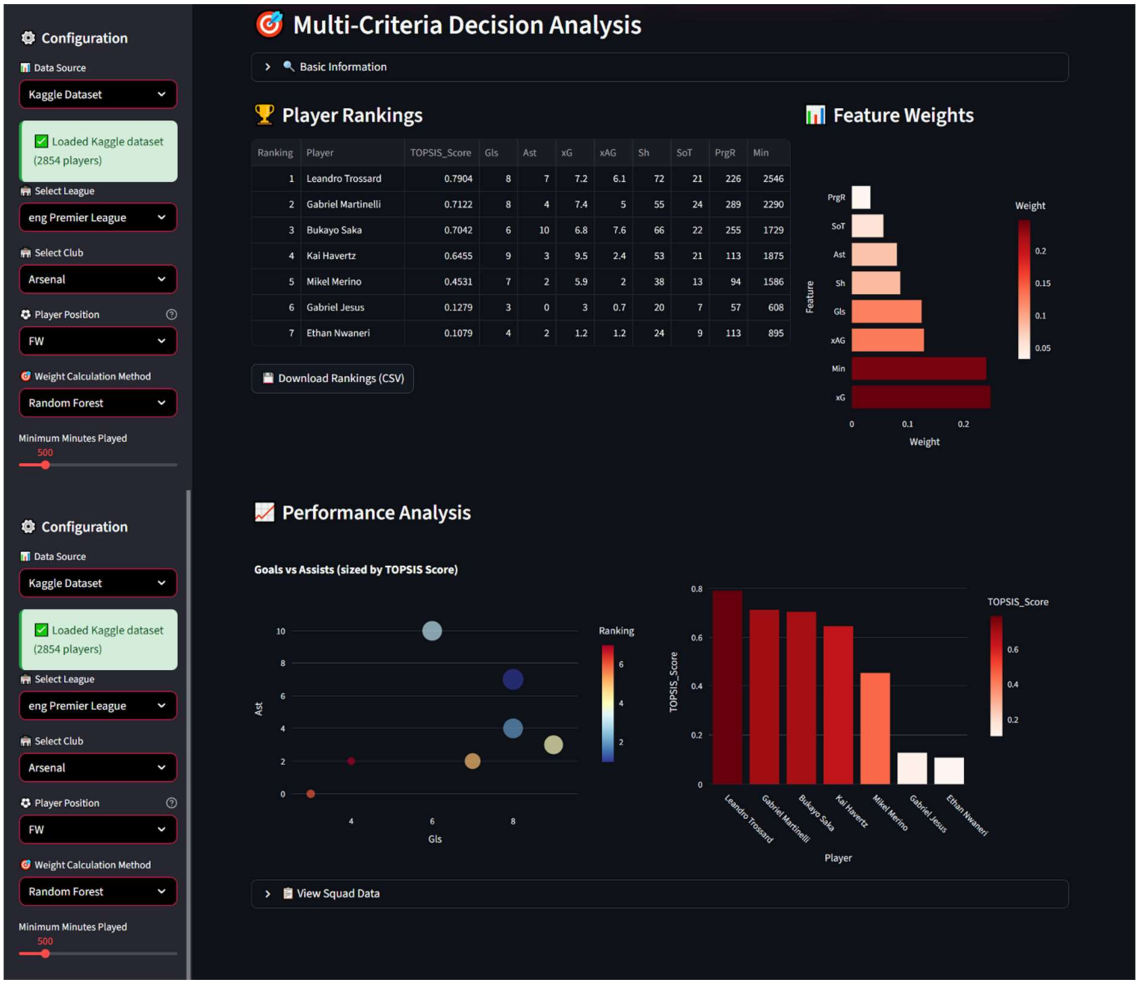

Figure 8. Impact of Alternative Criteria Settings on Arsenal Forwards: Player Rankings, Feature Weights, and Performance Analysis

In practice, criteria selection and weighting should be considered an adjustable parameter rather than a constant, and criteria relevance to the position and transparent benefit/cost perspective provide the most optimal, readable, and robust shortlists. Weighting schemes can significantly influence marginal decisions, therefore, the users should:

- Start from position-dependent default criteria.
- Inspect feature weights and their robustness (e.g., Kendall's т) upon every modification.
- Document the selected configuration along with the outcome to ensure transparency and reproducibility, thus future releases may incorporate expert feedback and automated criteria/weighting search, which further enhances its validity and applicability.

6.8. Decision Support System Design for Transparency and Usability (RQ6)

Transparency, usability, and reproducibility are key principles of the decision support system and is targeted at scouting departments and performance analysts. The framework enables the configuration of data sources, selection of leagues, clubs, player positions, weighting schemes, and filter options such as minimum number of minutes played, so no technical expertise is required.

Player rankings contain performance indicators and TOPSIS scores, and visualization of feature weights and performance is provided. The decision support system is intuitive to use and supports the decision-making process, thus rankings can be exported as CSV files for reporting purposes.

Weights for each evaluation criterion are presented transparently, and settings used to configure results are documented. It also provides reproducibility and confidence, because stability measures such as Kendall's τ indicate the variation in rankings when varying the configuration.

Several weighting strategies are provided, such as Random Forest, Entropy, and CRITIC, therefore criteria and parameters can be tuned. This enables to model various situations and personalized analysis depending on the strategy or tactic. The software is designed for end-users, such as scouts and coaches. It features a web-based, responsive user interface compatible with any device, so it is seamlessly integrated into work processes and can be used for player recruitment and performance assessment.

6.9. Summary of Findings and Implications for Practice

This chapter has presented a comprehensive analysis of the proposed decision support system for player selection in professional football. The multi-criteria approach demonstrated robustness and adaptability across different player positions, leagues, and clubs, effectively integrating diverse performance metrics into coherent rankings.

Key findings include the importance of selecting relevant criteria tailored to player roles, the impact of weighting methods on ranking stability, and the value of aligning system outputs with external benchmarks and expert assessments. Sensitivity analyses confirmed the system's resilience to variations in normalization and weighting, while highlighting the need for transparent and interpretable configurations.

From a practical perspective, the system offers a valuable tool for scouting and performance analysis teams, enabling evidence-based decision-making supported by clear visualizations and configurable parameters. Its design prioritizes usability and transparency, facilitating integration into existing workflows and promoting trust among users.

Future work involves incorporating expert judgements, additional data sources to assess intangible player attributes, and automated methods for optimal factor and weighting identification, because these enhancements will further improve the system's effectiveness, applicability, and acceptance among practitioners. These enhancements will further improve the system's effectiveness, applicability, and acceptance among practitioners, therefore leading to more widespread adoption and use of the system.

In summary, evidence-based decision support systems utilizing multiple factors can enhance player selection and contribute to more informed, equitable, and strategic decisions in football organizations, thus leading to improved performance and competitiveness, and therefore, they have the potential to revolutionize the way football organizations approach player selection.

7. Conclusions

7.1. Summary of aims and approach

The purpose of this research was to develop and validate an effective, data-driven player selection framework for elite European football, and a multi-criteria decision analysis (MCDA) model was designed to aggregate multiple performance indicators while accounting for positional differences (attacking, midfield, defensive, goalkeeper). Advanced weighting techniques including Random Forest, Entropy, and CRITIC were used to determine feature importance and derive robust TOPSIS-based player rankings, therefore allowing for a comprehensive evaluation of player performance.

Methodology consisted of three stages, so it was structured to ensure a thorough and systematic approach. Firstly, the quantitative data was cleaned and adjusted to make player statistics comparable across different leagues and teams, allowing for a fair evaluation of all players. Secondly, sensitivity analysis was performed to understand how changes in key design choices including metric selection, benefit/cost orientation, normalization, and weighting schemes on outcomes. Kendall's τ was used to measure how stable the rankings remained despite these changes. Lastly, external validation was performed by benchmarking shortlisted players against existing benchmarks (e.g., SoFIFA "Best Overall") and expert judgment to ensure practical applicability and validity, because this approach establishes a transparent, replicable, and position-specific methodology for evidence-based player selection.

7.2. Answers to the research questions

The study addressed six research questions regarding the design, robustness, and application of the decision tool, and the results indicated that:

- The multi-criteria approach aggregates difficult to interpret performance metrics into interpretable player rankings, thus providing a clear understanding of player performance.
- Position-specific criteria and weighting schemes increase the validity and interpretability of the rankings, because they allow for a more nuanced evaluation of player abilities.
- The decision framework is robust to variations in normalisation and weighting schemes; therefore, Random Forest provided the highest robustness.
- External benchmarks and expert judgement confirm the practical utility of the decision tool, so it can be relied upon for making informed decisions.
- Sensitivity analyses demonstrate the need for explicit, documented, and adjustable configurations, because this allows for greater flexibility and adaptability in the decision-making process.
- An intuitive, open-source design facilitates the adoption of the decision tool by scouting and performance departments, thus enabling them to make more effective use of the tool.

7.3. Contributions

The contribution to sports analytics of this project is a comprehensive, adaptable framework for evidence-based player selection and we refine our approach by combining multi-rule selection with machine learning for weighting. Our framework is adaptable to any tactical formation or player role, therefore validating the framework using real-world data and expert opinion demonstrates its effectiveness. We also gain insight from empirical results regarding how rule selection and weighting affects rankings; therefore, this facilitates development and refinement of novel decision support tools in sport.

7.4. Managerial implications and practical guidance

This project contributes to sports analytics by presenting a comprehensive and adaptable framework for evidence-based player selection, combining multi-criteria decision-making with machine learning-based weighting techniques. The framework is flexible enough to accommodate any tactical formation or player role. Validation using real-world data and expert opinions demonstrates its effectiveness and practical relevance. Insights gained from empirical results regarding the impact of criteria selection and weighting on player rankings provide valuable guidance for the development and refinement of future decision support tools in sports.

7.5. Limitations

Some limitations exist and the system requires quality, complete, and recent data. It does not capture all intangible attributes (such as leadership, off-the-ball movement, temperament) and contextual factors (such as tactics, opposition, game state) because past-game data may fail to identify emerging talent or recent form trends. User training may be required to optimally operate the system due to the multiple configurations and applications, therefore integration with existing club workflows may be challenging, particularly where data, IT policy, or software differs, thus requiring additional considerations.

7.6. Future work

Future research could focus on:

- Incorporating expert knowledge regarding aspects such as leadership, role, and strategy and combining quantitative and qualitative information.
- Expanding the scope of data used, including player trajectories, events, context (opponent, game state, role), and medical/physical data, which enhances situational awareness.
- Automating rule and weight derivation using meta-heuristics or Bayesian approaches, therefore reducing manual effort and ensuring consistency.
- Incorporating temporal aspects in player evaluation, so consider historical performance and current form, and utilize exponential smoothing, moving averages, or state-space models.
- Presenting confidence intervals for rankings and ratings, thus applying bootstrapping or Bayesian techniques, and provide intervals for risk assessment.
- Developing tools for decision-making regarding team composition and signings. Is essential for research to have a real impact and to be considered a success.

Adhere to constraints such as salary caps, homegrown quotas, and playing styles, and integer programming or evolutionary algorithms can be employed.

Provide transparency into the reasons behind rankings, so local explanations, hypothetical scenarios, and simulations can help coaches and scouts make sense of the rankings. Ease of use and maintenance, which means establishing data channels, tracking models and versions, recording parameters, and creating web interfaces for seamless integration with club systems.

Longitudinal analysis of outcomes, which assesses whether transfers pay off and players improve, and this validates the approach, allowing for iterative improvements.

7.7. Final statement

This dissertation demonstrates how a multi-factorial and data-driven approach can be applied to decision-making processes in football, and the proposed framework contributes to a more objective, transparent and strategic approach to player selection. It supports clubs in making informed decisions regarding player acquisitions and performance assessments, because it also provides an opportunity for improvements in scouting and coaching activities. Further research is required to incorporate additional data sources, such as expert knowledge, and refine predictive models, therefore this will contribute to sports management and competitive success in an increasingly data-intensive environment.

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Appendix A - Key player statistics and metrics from the 2024-2025 season Kaggle dataset.

Category	Variable	Description
Basic Player Information	Player	Player's name
	Nation	Player's nationality
	Pos	Playing position (FW, MF, DF, GK)
	Squad	Club or team name
	Comp	Competition or league name
	Age	Player's age
	Born	Year of birth
Playing Time & Appearances	MP	Matches played
	Starts	Matches started as a starter
	Min	Minutes played
	90s	Number of full 90-minute matches played
Attacking Stats	Gls	Goals scored
	Ast	Assists provided
	G+A	Goals plus assists
	xG	Expected goals (quality of scoring chances)
	xAG	Expected assists (quality of chances created)
	npxG	Non-penalty expected goals
	G-PK	Goals excluding penalties
	PK	Penalties scored
	PKatt	Penalties attempted
	Sh	Shots taken
	SoT	Shots on target
	SoT%	Percentage of shots on target
	FK	Free kicks taken
	Dist	Average shot distance
Passing & Creativity Stats	Cmp	Passes completed
	Att	Passes attempted

	Cmp%	Pass completion percentage
	KP	Key passes (passes leading to a shot)
	1/3	Passes into the attacking third
	PPA	Passes into the penalty area
	CrsPA	Crosses into the penalty area
	PrgP	Progressive passes (passes that move the ball significantly forward)
	PrgC	Progressive carries (ball carries moving the ball forward significantly)
	PrgR	Total progressive runs (progressive passes + carries)
	xA	Expected assists
Defensive Stats	Tkl	Total tackles
	TklW	Tackles won
	Blocks	Blocks made
	Int	Interceptions
	Tkl+Int	Combined tackles and interceptions
	Clr	Clearances
	Err	Errors leading to goals
	CrdY	Yellow cards
	CrdR	Red cards
Possession & Ball Control	Touches	Total touches of the ball
	Carries	Total ball carries
	Mis	Miscontrols (loss of ball control)
	Dis	Times dispossessed
	Recov	Ball recoveries
Goalkeeping Stats (GK only)	GA	Goals conceded
	Saves	Saves made
	Save%	Save percentage
	CS	Clean sheets
	CS%	Clean sheet percentage
	PKA	Penalties faced

	PKsv	Penalty saves
Advanced & Derived Stats	G+A-PK	Goals plus assists excluding penalties
	xG+xAG	Sum of expected goals and expected assists
	G-xG	Difference between goals scored and expected goals (over or underperformance)
	np:G-xG	Difference between non-penalty goals and non-penalty expected goals
	Sh/90	Shots per 90 minutes
	SoT/90	Shots on target per 90 minutes
	G/Sh	Goals per shot
	G/SoT	Goals per shot on target
	npxG/Sh	Non-penalty expected goals per shot
	Rk_stats_shooting	Ranking within shooting statistics
	Nation_stats_shooting	National average or ranking in shooting stats
	Pos_stats_shooting	Positional average or ranking in shooting stats
	Comp_stats_shooting	Competition average or ranking in shooting stats
	Age_stats_shooting	Age group average or ranking in shooting stats
	Born_stats_shooting	Birth year group average or ranking in shooting stats
	90s_stats_shooting	Average or ranking of 90-minute intervals in shooting stats
	Rk_stats_passing	Ranking within passing statistics
	Nation_stats_passing	National average or ranking in passing stats
	Pos_stats_passing	Positional average or ranking in passing stats
	Comp_stats_passing	Competition average or ranking in passing stats
	Age_stats_passing	Age group average or ranking in passing stats

	Born_stats_passing	Birth year group average or ranking in passing stats
	90s_stats_passing	Average or ranking of 90-minute intervals in passing stats
	Rk_stats_defense	Ranking within defensive statistics
	Nation_stats_defense	National average or ranking in defensive statistics
	Pos_stats_defense	Positional average or ranking in defensive statistics
	Comp_stats_defense	Competition average or ranking in defensive statistics
	Age_stats_defense	Age group average or ranking in defensive statistics
	Born_stats_defense	Birth year group average or ranking in defensive statistics
	90s_stats_defense	Average or ranking of 90-minute intervals in defensive statistics

Appendix B – Python Code

The Python code was developed and executed in the PyCharm integrated development environment (IDE), version 2025.2.0.1, with Python version 3.10. The libraries listed in the appendix were used to ensure reproducibility of the analysis.

```
# Appendix B: Installation of required libraries
# Run these commands in the terminal to install the libraries:
# pip install streamlit pandas numpy scikit-learn plotly kagglehub scipy

# Finally, the user must run the Streamlit application on localhost, open the terminal (or
console)
# in the folder where the Python file (mcdm_tool_2024_2025.py) is located and type this
command:
# streamlit run mcdm_tool_2024_2025.py

import streamlit as st
import pandas as pd
import numpy as np
from sklearn.ensemble import RandomForestRegressor
import plotly.express as px
import warnings
import kagglehub
from sklearn.metrics import ndcg_score
from scipy.stats import kendalltau

warnings.filterwarnings('ignore')

# ===== CONFIGURATION =====
st.set_page_config(
    page_title="Football DSS - Player Selection Optimization",
    page_icon="⚽",
    layout="wide",
    initial_sidebar_state="expanded"
)

# ===== IMPROVED CSS STYLES =====
st.markdown("""
<style>
    .main-header {
        background: linear-gradient(90deg, #AC1E44 0%, #000000 100%);
        padding: 2rem;
        border-radius: 10px;
        text-align: center;
        margin-bottom: 2rem;
        color: white;
    }
    .metric-card {
        background: linear-gradient(135deg, #AC1E44 0%, #8B1538 100%);
        padding: 1.5rem;
        border-radius: 15px;
        text-align: center;
        color: white;
        box-shadow: 0 8px 32px rgba(172, 30, 68, 0.3);
        border: 1px solid rgba(255, 255, 255, 0.1);
```

```
    }
    .metric-card h3 {
        margin: 0;
        font-size: 1rem;
        opacity: 0.9;
        font-weight: 500;
    }
    .metric-card h2 {
        margin: 0.5rem 0 0 0;
        font-size: 2.5rem;
        font-weight: bold;
    }
    .stSelectbox > div > div {
        background-color: #000000 !important;
        color: #ffffff !important;
        border: 2px solid #AC1E44;
        border-radius: 10px;
    }
    .stSelectbox > div > div > div {
        background-color: #000000 !important;
        color: #ffffff !important;
    }
    .debug-info {
        background-color: #1e1e1e;
        color: #00ff00;
        padding: 1rem;
        border-radius: 8px;
        font-family: 'Courier New', monospace;
        font-size: 0.85rem;
    }
    .success-msg {
        background-color: #d4edda;
        color: #155724;
        padding: 1rem;
        border-radius: 8px;
        border-left: 4px solid #28a745;
    }
</style>
""", unsafe_allow_html=True)


# ===== MCDM FUNCTIONS =====
def entropy_weights(data):
    data_clean = np.abs(data) + 1e-10
    data_clean = np.nan_to_num(data_clean, nan=1e-10, posinf=1e10, neginf=1e-10)
    data_norm = data_clean / np.sum(data_clean, axis=0)
    data_norm = np.clip(data_norm, 1e-10, 1.0)
    entropy = -np.sum(data_norm * np.log(data_norm), axis=0) / np.log(len(data_norm))
    entropy = np.nan_to_num(entropy, nan=0.5)
    weights = (1 - entropy) / np.sum(1 - entropy)
    return weights / np.sum(weights)


def critic_weights(data):
    data_clean = np.nan_to_num(data, nan=0, posinf=1e10, neginf=-1e10)
```

```
    std_dev = np.std(data_clean, axis=0)
    std_dev = np.where(std_dev == 0, 1e-10, std_dev)
    corr_matrix = np.corrcoef(data_clean.T)
    corr_matrix = np.nan_to_num(corr_matrix, nan=0)
    conflict = np.sum(1 - np.abs(corr_matrix), axis=1)
    critic_values = std_dev * conflict
    weights = critic_values / np.sum(critic_values)
    return weights / np.sum(weights)


def random_forest_weights(data, target_col='Gls'):
    data_clean = data.fillna(0)
    if target_col not in data_clean.columns:
        return np.ones(len(data_clean.columns)) / len(data_clean.columns)
    X = data_clean.drop(columns=[target_col])
    y = data_clean[target_col]
    rf = RandomForestRegressor(n_estimators=100, random_state=42)
    rf.fit(X, y)
    importances = rf.feature_importances_
    weights = np.zeros(len(data_clean.columns))
    feature_indices = [i for i, col in enumerate(data_clean.columns) if col != target_col]
    for i, importance in enumerate(importances):
        weights[feature_indices[i]] = importance
    weights[data_clean.columns.get_loc(target_col)] = np.mean(importances)
    return weights / np.sum(weights)


def topsis_ranking(data, weights, is_beneficial=None):
    # Replace NaN and infinite values
    data_clean = np.nan_to_num(data, nan=0, posinf=1e10, neginf=-1e10)

    # Eliminate constant columns to avoid division by zero
    std_dev = np.std(data_clean, axis=0)
    non_const_cols = std_dev > 1e-10
    data_clean = data_clean[:, non_const_cols]
    weights = weights[non_const_cols]
    if is_beneficial is not None:
        is_beneficial = np.array(is_beneficial)[non_const_cols]
    else:
        is_beneficial = [True] * data_clean.shape[1]

    weights = weights / np.sum(weights)
    norm_data = data_clean / np.sqrt((data_clean ** 2).sum(axis=0))
    weighted_data = norm_data * weights
    ideal_pos = np.where(is_beneficial, weighted_data.max(axis=0),
weighted_data.min(axis=0))
    ideal_neg = np.where(is_beneficial, weighted_data.min(axis=0),
weighted_data.max(axis=0))
    dist_pos = np.sqrt(((weighted_data - ideal_pos) ** 2).sum(axis=1))
    dist_neg = np.sqrt(((weighted_data - ideal_neg) ** 2).sum(axis=1))
    return dist_neg / (dist_pos + dist_neg)


# ===== MAIN =====
def main():
```

```
    ndcg = None
    tau = None
    st.markdown("""
    <div class="main-header">
        <h1>⚽ Football DSS</h1>
        <h3>Decision Support System for Player Selection Optimization</h3>
        <p>Integrating Machine Learning & Multi-Criteria Decision Analysis</p>
    </div>
    """, unsafe_allow_html=True)

    with st.sidebar:
        st.markdown("## ⚙️ Configuration")

        data_source = st.selectbox("📊 Data Source",
                                   ["Kaggle Dataset", "AC Milan Sample Data"])

        # NEW FILTERS
        selected_league, selected_club = "All Leagues", "All Clubs"
        if data_source == "Kaggle Dataset":
            path = kagglehub.dataset_download("hubertsidorowicz/football-players-stats-
2024-2025")
            df = pd.read_csv(f'{path}/players_data-2024_2025.csv')
            st.markdown(f'<div class="success-msg">✅ Loaded Kaggle dataset ({len(df)}
players)</div>',
                        unsafe_allow_html=True)

            # League Filter
            leagues = ["All Leagues"] + sorted(df['Comp'].dropna().unique().tolist())
            selected_league = st.selectbox("🏟️ Select League", leagues)
            if selected_league != "All Leagues":
                df = df[df['Comp'] == selected_league]

            # Club Filter
            clubs = ["All Clubs"]
            if selected_league != "All Leagues":
                clubs += sorted(df['Squad'].dropna().unique().tolist())
            selected_club = st.selectbox("🏟️ Select Club", clubs)
            if selected_club != "All Clubs":
                df = df[df['Squad'] == selected_club]

        else:  # Sample AC Milan
            sample_data = {
                'Player': ['Mike Maignan', 'Theo Hernández', 'Fikayo Tomori', 'Pierre Kalulu',
                           'Davide Calabria', 'Sandro Tonali', 'Ismaël Bennacer', 'Rafael Leão',
                           'Brahim Díaz', 'Olivier Giroud', 'Ante Rebić'],
                'Pos': ['GK', 'DF', 'DF', 'DF', 'DF', 'MF', 'MF', 'FW', 'MF', 'FW', 'FW'],
                'Age': [28, 26, 25, 24, 27, 23, 26, 24, 24, 37, 30],
                'MP': [38, 35, 31, 20, 29, 34, 32, 34, 28, 31, 25],
                'Min': [3420, 3150, 2790, 1800, 2610, 3060, 2880, 3060, 2520, 2790, 2250],
                'Gls': [0, 3, 2, 1, 2, 8, 3, 11, 6, 11, 4],
                'Ast': [0, 8, 1, 0, 3, 7, 4, 10, 4, 4, 2],
                'xG': [0.0, 2.1, 1.8, 0.8, 1.2, 6.2, 2.1, 8.9, 4.2, 9.8, 3.1],
                'xAG': [0.0, 6.8, 0.9, 0.2, 2.1, 5.8, 3.2, 8.1, 3.8, 2.9, 1.8],
                'PrgC': [12, 89, 45, 23, 67, 156, 134, 178, 98, 67, 45],
```

```
            'PrgP': [245, 234, 178, 89, 156, 289, 267, 198, 167, 134, 98]
        }
        df = pd.DataFrame(sample_data)

    # Position filter
    position_options = ["All Positions", "FW", "MF", "DF", "GK"]
    position_filter = st.selectbox("⚽ Player Position",
                                   position_options,
                                   help="Filter players by role")

    weight_method = st.selectbox("🎯 Weight Calculation Method",
                                 ["Random Forest", "Entropy", "CRITIC"])
    min_minutes = st.slider("Minimum Minutes Played", 0, 3000, 500)

# ==== GLOBAL FILTERS ====
if 'Min' in df.columns:
    df = df[df['Min'] >= min_minutes]

if position_filter != "All Positions" and 'Pos' in df.columns:
    def expand_positions(pos):
        if pd.isna(pos):
            return []
        return [p.strip() for p in pos.split(",")]
    df = df[df['Pos'].apply(lambda x: position_filter in expand_positions(x))]

# ===== KEY METRICS =====
st.markdown("<br>", unsafe_allow_html=True)

col1, col2, col3, col4 = st.columns(4)
with col1:
    st.markdown(f"<div class='metric-card'><h3>👥 Total Players</h3><h2>{len(df)}</h2></div>", unsafe_allow_html=True)
with col2:
    st.markdown(f"<div class='metric-card'><h3>📅 Average Age</h3><h2>{df['Age'].mean():.1f}</h2></div>", unsafe_allow_html=True)
with col3:
    st.markdown(f"<div class='metric-card'><h3>⚽ Total Goals</h3><h2>{df['Gls'].sum()}</h2></div>", unsafe_allow_html=True)
with col4:
    st.markdown(f"<div class='metric-card'><h3>🎯 Total Assists</h3><h2>{df['Ast'].sum()}</h2></div>", unsafe_allow_html=True)

# ===== MCDM =====
st.markdown("## 🎯 Multi-Criteria Decision Analysis")

numeric_cols = df.select_dtypes(include=[np.number]).columns.tolist()

if position_filter == "FW":
    preferred_cols = ['Gls', 'Ast', 'xG', 'xAG', 'Sh', 'SoT', 'PrgR', 'Min']
elif position_filter == "MF":
    preferred_cols = ['Ast', 'KP', 'xAG', 'PrgP', 'PrgC', 'Gls', 'Min']
elif position_filter == "DF":
    preferred_cols = ['Tkl', 'Int', 'Clr', 'Blocks', 'PrgP', 'CrdY', 'CrdR', 'Min']
```

```
elif position_filter == "GK":
    preferred_cols = ['GA90', 'Save%', 'CS%', 'PKsv', 'Min']
else:
    preferred_cols = ['Gls', 'Ast', 'xG', 'xAG', 'Min', 'PrgC', 'PrgP']

analysis_cols = [c for c in preferred_cols if c in numeric_cols]
if len(analysis_cols) < 3:
    analysis_cols = numeric_cols[:min(8, len(numeric_cols))]

if len(analysis_cols) >= 3:
    analysis_df = df[analysis_cols + ['Player']].copy()
    analysis_data = analysis_df[analysis_cols].fillna(0)

    if weight_method == "Random Forest":
        weights = random_forest_weights(analysis_data)
    elif weight_method == "Entropy":
        weights = entropy_weights(analysis_data.values)
    else:
        weights = critic_weights(analysis_data.values)

    # criteria to be minimized
    minimize_cols = ['CrdY', 'CrdR', 'GA90']
    is_beneficial = [False if col in minimize_cols else True for col in analysis_cols]

    # === Generate TOPSIS scores ===
    topsis_scores = topsis_ranking(analysis_data.values, weights, is_beneficial)

    results_df = analysis_df.copy()
    results_df['TOPSIS_Score'] = pd.Series(topsis_scores, index=analysis_df.index)

    # Replace NaN by 0 before slotting
    results_df['TOPSIS_Score'] = results_df['TOPSIS_Score'].fillna(0)

    results_df['Ranking'] = results_df['TOPSIS_Score'].rank(ascending=False).astype(int)
    results_df = results_df.sort_values('Ranking')

    # Define reference metric for true_relevance by position
    if position_filter in ['FW', 'MF']:
        relevance_col = 'Gls'
    elif position_filter == 'DF':
        # Use a representative defensive metric
        relevance_col = 'Tkl' if 'Tkl' in results_df.columns else None
    elif position_filter == 'GK':
        relevance_col = 'Save%' if 'Save%' in results_df.columns else None
    else:
        relevance_col = 'Gls'  # Default fallback

    if relevance_col and relevance_col in results_df.columns:
        true_relevance = results_df[relevance_col].values
    else:
        st.error(f"Reference metric '{relevance_col}' not found for position {position_filter}.")
        true_relevance = np.zeros(len(results_df))

    pred_scores = results_df['TOPSIS_Score'].values
```

```
    if len(true_relevance) < 2 or true_relevance.max() == 0:
        ndcg = 0
        tau = 0
    else:
        true_relevance_norm = true_relevance / true_relevance.max()
        ndcg = ndcg_score([true_relevance_norm], [pred_scores])
        tau, _ = kendalltau(true_relevance, pred_scores)
    with st.expander("🔍 Basic Information"):
        ndcg_display = f"{ndcg:.4f}" if ndcg is not None else "N/A"
        tau_display = f"{tau:.4f}" if tau is not None else "N/A"
        st.markdown(f"""
        <div class="debug-info">
        📊 Dataset Shape: {df.shape}<br>
        📋 Columns: {len(df.columns)}<br>
        🎯 Weight Method: {weight_method}<br>
        📈 Data Source: {data_source}<br>
        🔢 Players after filter: {len(df)}<br>
        📊 NDCG Score: {ndcg_display}<br>
        📊 Kendall's Tau: {tau_display}
        </div>
        """, unsafe_allow_html=True)
    # Add Squad and Comp if they exist in original df
    if 'Squad' in df.columns:
        results_df['Squad'] = df.loc[results_df.index, 'Squad']
    if 'Comp' in df.columns:
        results_df['Comp'] = df.loc[results_df.index, 'Comp']

    # ===== RESULTS =====
    col1, col2 = st.columns([2, 1])
    with col1:
        st.markdown("### 🏆 Player Rankings")

        display_cols = ['Ranking', 'Player']

        # Show 'Squad' only if no specific club is selected
        if 'Squad' in results_df.columns and selected_club == "All Clubs":
            display_cols.append('Squad')

            # Show 'Comp' only if "All Leagues" is selected
        if 'Comp' in results_df.columns and selected_league == "All Leagues":
            display_cols.append('Comp')
        display_cols += ['TOPSIS_Score'] + analysis_cols

        st.dataframe(results_df[display_cols].head(15),
                     use_container_width=True, hide_index=True)
        # ==== Export results ====
        csv_export = results_df.to_csv(index=False).encode('utf-8')
        st.download_button(
            label="💾 Download Rankings (CSV)",
            data=csv_export,
            file_name="player_rankings.csv",
            mime="text/csv"
        )
```

```
        with col2:
            st.markdown("### 📊 Feature Weights")
            weights_df = pd.DataFrame({'Feature': analysis_cols,
                                       'Weight': weights[:len(analysis_cols)]}).sort_values('Weight',
ascending=False)
            fig_weights = px.bar(weights_df, x='Weight', y='Feature', orientation='h',
                                 color='Weight', color_continuous_scale='Reds')
            st.plotly_chart(fig_weights, use_container_width=True)

        # ===== VISUALIZATIONS =====
        st.markdown("### 📈 Performance Analysis")
        col1, col2 = st.columns(2)
        if 'Gls' in results_df and 'Ast' in results_df:
            fig_scatter = px.scatter(results_df.head(20),
                                     x='Gls', y='Ast',
                                     size='TOPSIS_Score', color='Ranking',
                                     hover_name='Player',
                                     title="Goals vs Assists (sized by TOPSIS Score)",
                                     color_continuous_scale='RdYlBu_r')
            col1.plotly_chart(fig_scatter, use_container_width=True)

        top10 = results_df.head(10)
        fig_top = px.bar(top10, x='Player', y='TOPSIS_Score',
                         color='TOPSIS_Score', color_continuous_scale='Reds')
        fig_top.update_xaxes(tickangle=45)
        col2.plotly_chart(fig_top, use_container_width=True)

        # ===== VIEW COMPLETE DATASET =====
        with st.expander("📋 View Squad Data"):
            st.dataframe(df, use_container_width=True)

    else:
        st.error("❌ Not enough numeric columns for analysis.")


if __name__ == "__main__":
    main()
```